COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE 010 - Data Structures and Algorithms**	**SECTION:**	**CPE21S2**
DATE PERFORMED:	**07-10-26**	**DATE SUBMITTED:**	**07-15-26**
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ACTIVITY NO. 1 **Hands-on Activity 1.1 Basic C++ Programming**

1. PROCEDURE OUTPUT

ILO A

Sections	Answer

Header File Declaration Section	HOA1 > ILOA > ILOA.cpp > main()
	`#include<iostream>`
	`//used for typeid function that checks the data type`
	`#include<typeinfo>`

Global Declaration Section	

Class Declaration and Method Definition Section	`//Create a function that will take two numbers and display the sum`
	`int sumOfTwoNum(int fNum, int sNum);`
	
	`//Create a function that will return whether variable A is greater than variable B`
	`int greaterThan(int varA, int varB);`
	
	`//Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.`
	`template <typename T> T logicOperations(T fVal, T sVal);`

Main Function	`int main(){`
	
	`std::cout << "The sum of two numbers is: "<< sumOfTwoNum(6,7) << std::endl;`
	
	`if(greaterThan(7,5) == 1){`
	`std::cout << "\nfirst variable is greater than the second variable\n" << std::endl;`
	`}else{`
	`std::cout << "\nfirst variable is not greater than the second variable\n" << std::endl;`
	`}`
	
	`if(logicOperations(true,true)){`
	`std::cout << "\nSuccess!" << std::endl;`
	`}else{`
	`std::cout << "Failed. The given values are not boolean values \n";`
	`}`
	
	`return 0;`
	`}`

Method Definition	

```cpp
int sumOfTwoNum(int fNum, int sNum){

    int sum = fNum + sNum;

    return sum;
}

int greaterThan(int varA, int varB){

    int answer = varA > varB;

    if(answer == 1){
        return 1;
    }else{
        return 0;
    }
}

template <typename T> T logicOperations(T fVal, T sVal){
    if(typeid(fVal) != typeid(bool) && typeid(sVal)!= typeid(bool)){
        return false;
    }

    bool And = fVal && sVal;
    bool Or = fVal || sVal;
    bool Not = !(fVal && sVal);

    std::cout <<"Result of AND operato: "<< And << std::endl;
    std::cout << "Result of OR operator: " << Or << std::endl;
    std::cout << "Result of NOT operator: " << Not << std::endl;

    return true;
}
```

Table 1. C++ Structure Code for Answer in ILO A

```
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\ILOA> cd "c:\Users\User\OneDrive\Documents\CPE010\HOA1\ILOA\" ;
The sum of two numbers is: 13

first variable is greater than the second variable

Result of AND operato: 1
Result of OR operator: 1
Result of NOT operator: 0

Success!
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\ILOA>
```

```
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\ILOA> cd "c:\Users\User\OneDrive\Documents\CPE010\HOA1\ILOA\" ;
The sum of two numbers is: 13

first variable is greater than the second variable

Failed. The given values are not boolean values
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\ILOA>
```

Observations

For the first two problems, it's just proper function prototype and definition, and solving basic problems. For the third problem, I had to include the typeinfo library for the typeid function which is used to determine the type of a variable and I also created a template so that the function can work with any data type. I also used the logic operators in c++ which was taught before.

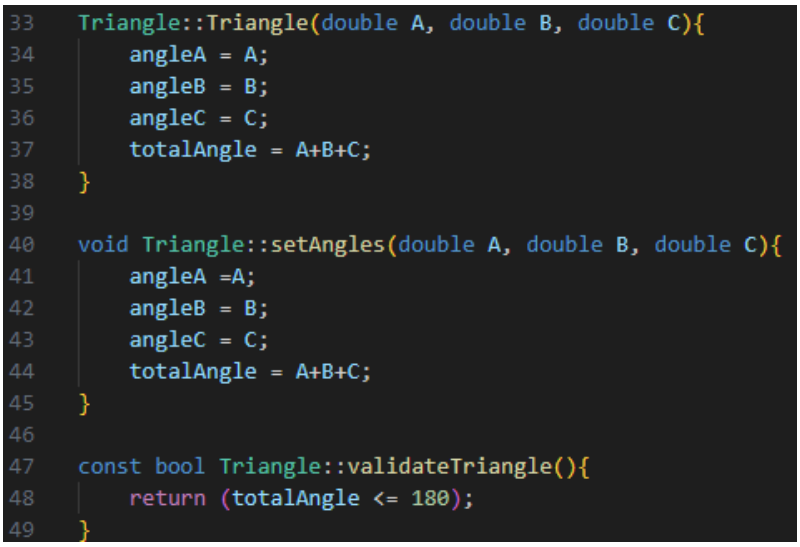
Table 2.ILO A Output Observations and Comments

ILO B

Sections	Answer
Header File Declaration Section	HOA1 > ILO B > main.cpp > main() `#include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	`class Triangle{`
	`private:`
	`double totalAngle, angleA, angleB, angleC;`
	`public:`
	`Triangle(double A, double B, double C);`
	`void setAngles(double A, double B, double C);`
	`const bool validateTriangle();`
	`};`

Main Function	`int main(){`
	`Triangle set1(40, 30, 110);`
	`if(set1.validateTriangle()){`
	`std::cout << "\nThe shape is a valid triangle.\n";`
	`}else{`
	`std::cout << "\nThe shape is NOT a valid triangle.\n";`
	`}`
	`set1.setAngles(40, 50, 110);`
	`if(set1.validateTriangle()){`
	`std::cout << "The shape is a valid triangle.\n\n";`
	`}else{`
	`std::cout << "The shape is NOT a valid triangle.\n\n";`
	`}`
	`return 0;`
	`}`

Method Definition	

```
33  Triangle::Triangle(double A, double B, double C){
34      angleA = A;
35      angleB = B;
36      angleC = C;
37      totalAngle = A+B+C;
38  }
39
40  void Triangle::setAngles(double A, double B, double C){
41      angleA =A;
42      angleB = B;
43      angleC = C;
44      totalAngle = A+B+C;
45  }
46
47  const bool Triangle::validateTriangle(){
48      return (totalAngle <= 180);
49  }
```

Table 3. C++ Structure Code for Answer in ILO B

```
PS C:\Users\TIPQC\Documents\CPE010 - Bandigan\HOA1\ILO A> cd "c:\Users\TIPQC\Documents\CPE010 - Bandigan\HOA1\ILO B\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }

The shape is a valid triangle.
The shape is NOT a valid triangle.
```

Observations

This is my first time making a class in c++. It is very similar to python with its private variables, constructor, and public methods, of course you make the declaration before main and definitions after main. To create an object you first type the class name and then the object variable with the proper arguments needed. To call for the member functions you need to use the dot operator.

Table 4. ILO B Output Observations and Comments

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

HOA1 > Supplementary > ILOC > ILOC.cpp > numberSwapper()

```cpp
#include <iostream>

//Create a C++ program to swap the two numbers in different variables.
void numberSwapper();

//global variables
int  varA = 6;
int varB = 9;

int main(){

    std::cout << "The value of the variables: \n" ;
    std::cout << "Variable A: " << varA << std::endl;
    std::cout << "Variable B: " << varB << std::endl;

    numberSwapper();

    std::cout << "\nThe value of the variables after swapping: \n";
    std::cout << "Variable A: " << varA << std::endl;
    std::cout << "Variable B: " << varB << std::endl;

    return 0;
}

void numberSwapper(){
    int temp;
    temp = varA;
    varA = varB;
    varB = temp;
}
```

Figure 1. Code of Supplementary Activity 1

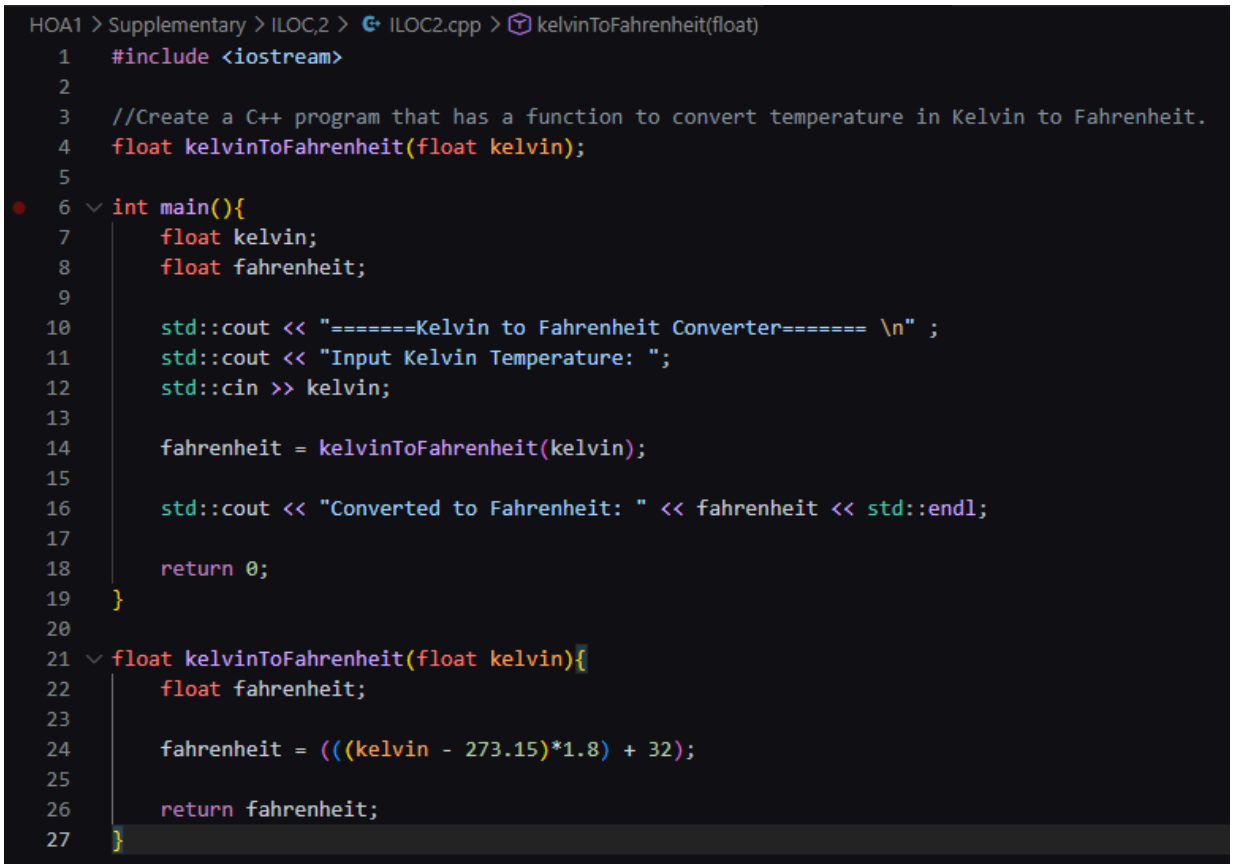
```
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOB Mod> cd "c:\Users\Use
The value of the variables:
Variable A: 6
Variable B: 9

The value of the variables after swapping:
Variable A: 9
Variable B: 6
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOC>
```

Figure 2. Output of Supplementary Activity 1

Analysis

In order to swap two numbers in different variables I just created a temporary variable and assigned it to the first variable and assigned that variable to the second variable and assigned that variable to the temporary variable. I declared it as global variables just so that the void function and main can access it.



```
HOA1 > Supplementary > ILOC,2 > ILOC2.cpp > kelvinToFahrenheit(float)
#include <iostream>

//Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.
float kelvinToFahrenheit(float kelvin);

int main(){
    float kelvin;
    float fahrenheit;

    std::cout << "=======Kelvin to Fahrenheit Converter======= \n" ;
    std::cout << "Input Kelvin Temperature: ";
    std::cin >> kelvin;

    fahrenheit = kelvinToFahrenheit(kelvin);

    std::cout << "Converted to Fahrenheit: " << fahrenheit << std::endl;

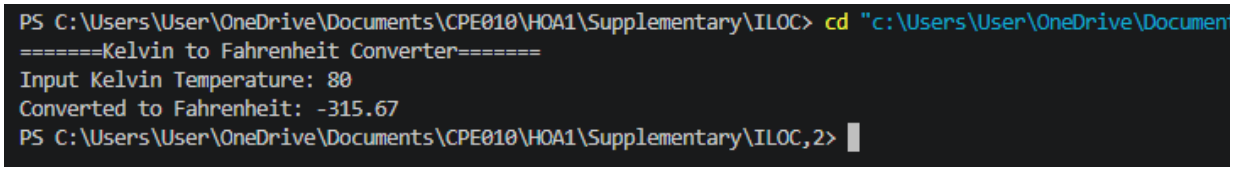
    return 0;
}

float kelvinToFahrenheit(float kelvin){
    float fahrenheit;

    fahrenheit = (((kelvin - 273.15)*1.8) + 32);

    return fahrenheit;
}
```

Figure 3. Code of Supplementary Activity 2



```
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOC> cd "c:\Users\User\OneDrive\Documen
=======Kelvin to Fahrenheit Converter=======
Input Kelvin Temperature: 80
Converted to Fahrenheit: -315.67
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOC,2>
```

Figure 4. Output of Supplementary Activity 2

Analysis

I first created a function prototype that takes and returns a float variable. The function takes in the user input kelvin variable and just converts it into fahrenheit using the formula and returns that variable.

HOA1 > Supplementary > ILOC,3 > ILOC3.cpp > main()

```cpp
#include <iostream>
//for square root and power
#include <cmath>

//Create a C++ program that has a function that will calculate the distance between two points.
float distanceCalculator(float x1, float y1, float x2, float y2);

int main(){

    float x1, y1, x2,y2,distance;

    std::cout << "=======Distance Calculator======= \n";
    std::cout << "Enter Coordinate of first point: \n";
    std::cout << "x1: ";
    std::cin >> x1;
    std::cout << "y1: ";
    std::cin >> y1;

    std::cout << "\nEnter Coordinate of second point: \n";
    std::cout << "x2: ";
    std::cin >> x2;
    std::cout << "y2: ";
    std::cin >> y2;

    distance = distanceCalculator(x1,y1,x2,y2);

    std::cout << "\nThe distance between two points is: " << distance << std::endl;

    return 0;
}

float distanceCalculator(float x1, float y1, float x2, float y2){
    float distance;

    distance = sqrt((pow((x2-x1),2)) + (pow((y2-y1),2)));

    return distance;
}
```

Figure 5.Code of Supplementary Activity 3

```
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOC,3> cd "c:\Users\User\OneDrive\Do
=======Distance Calculator=======
Enter Coordinate of first point:
x1: 12
y1: 6

Enter Coordinate of second point:
x2: 20
y2: 5

The distance between two points is: 8.06226
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOC,3>
```

Figure 6. Output of Supplementary Activity 3

Analysis

I first included the cmath library for its sqrt() and pow() functions that will be used for the formula for getting the distance between two points. After that, I created a float function prototype that takes in four float variables for the coordinates of the two points. I asked for coordinates of the two points from the user and just called the function with the four variables. Inside the function definition it just inserts the arguments inside the formula for getting the distance between two points and assigns it to a variable and finally returns that variable.

HOA1 > Supplementary > ILOB Mod > ILOB.cpp > ...

```cpp
#include <iostream>

class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
        //A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'
        int triangleType();
};

//A function to compute for the area of a triangle
float triangleArea(float base, float height);

//A function to compute for the perimeter of a triangle
float trianglePerimeter(float side1, float side2, float side3);

int main(){
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "\nThe shape is a valid triangle. \n";
    }else{
        std::cout << "\nThe shape is NOT a valid triangle. \n";
    }

    set1.setAngles(40,40,100);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle. \n\n";
    }else{
        std::cout << "The shape is NOT a valid triangle. \n\n";
    }

    std::cout << "The area of a triangle where it's base is equal to 10 and height is equal to 8: "<< triangleArea(10,8) << std::endl;

    std::cout << "\nThe perimeter of a triangle where the length of it's sides are all 8: " << trianglePerimeter(8,8,8) << std::endl;

    if(set1.triangleType() == 1){
        std::cout << "The given triangle is obtuse-angled.";
    }else if(set1.triangleType() == 2){
        std::cout << "The given triangle is acute-angled.";
    }else if(set1.triangleType() == 3){
        std::cout << "The given triangle is a right triangle.";
    }else{
        std::cout << "The given shape is NOT a valid triangle.";
    }

    return 0;
}
```

Figure 7. Function Prototype and Main code of Supplementary Activity 4

```
52
53  Triangle::Triangle(double A, double B, double C){
54      angleA = A;
55      angleB = B;
56      angleC = C;
57      totalAngle = A + B + C;
58  }
59  void Triangle::setAngles(double A, double B, double C){
60      angleA = A;
61      angleB = B;
62      angleC = C;
63      totalAngle = A + B + C;
64  }
65
66  const bool Triangle::validateTriangle(){
67      return (totalAngle <=180);
68  }
69
70  float triangleArea(float base, float height){
71      float area;
72
73      area = 0.5*(base*height);
74
75      return area;
76  }
77
78  float trianglePerimeter(float side1, float side2, float side3){
79      return side1 + side2 + side3;
80  }
81
82  int Triangle::triangleType(){
83      double angles[3] = {angleA, angleB, angleC};
84      if(totalAngle <=180){
85         for(int i = 0;i <=3; i++){
86          if(angles[i] > 90){
87              return 1;
88          }
89          else if(angles[i] == 90){
90              return 3;
91          }
92      }
93      return 2;
94      }
95      return 0;
96  }
```

Figure 8. Function Definitions of Supplementary Activity 4

```
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOB Mod> cd "c:\Users\User\OneDrive\Documents\CP

The shape is a valid triangle.
The shape is a valid triangle.

The area of a triangle where it's base is equal to 10 and height is equal to 8: 40

The perimeter of a triangle where the length of it's sides are all 8: 24
The given triangle is obtuse-angled.
PS C:\Users\User\OneDrive\Documents\CPE010\HOA1\Supplementary\ILOB Mod>
```

Figure 9. Output of Supplementary Activity 4

Analysis

A-B. To compute the area and perimeter of the triangle it just takes the base, height, and sides from the user and plug it inside the formula. All this process is inside the float function.

C. To determine the type of the given triangle I created a member function (int triangleType()) inside the Triangle class. Inside the function definition it first takes the private variables of the class and puts it into an array so that I can check every angle. After that, I first checked if the given shape is a valid triangle and then used a for loop to check every angle. Finally, it returns an int variable inside main where it will be used in an if… else if statement and prints out the proper type of the given triangle.

Conclusion

In this activity I was able to remember and use my knowledge of c++ from a previous course like function prototypes, proper c++ code structure, mathematical operations, etc. and combine these with new knowledge like templates and typeid that checks data types. Moreover, I was able to utilize what I learned about OOP by implementing classes and methods in the code. Overall, I was able to review what I know about basic c++ programming and was able to solve different problems.

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